ALICE L WALTON SCHOOL OF MEDICINE

Essential Thrombocythemia

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Adapted from: Essential Thrombocythemia

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of essential thrombocythemia.
2. Explain the pathophysiology of essential thrombocythemia.
3. Interpret clinical findings to properly diagnose essential thrombocythemia.
4. Outline the management plan of essential thrombocythemia.

CASE CONNECTION

SS is a 68-year-old female who is referred by her neurologist after initial evaluation of new-onset headaches and occasional blurry vision. Her ophthalmologic evaluation and head CT scan are normal. "They're not sure what is causing these symptoms. I've had several tests already." SS has no relevant medical history. She does not drink alcohol or smoke. Her vitals are within normal limits, and a physical exam is remarkable only for splenomegaly. As you review the record, you are startled by one of the results of her complete blood count (CBC), a very high platelet count.

What is the diagnosis? How will you describe SS's condition to her? Consider your answers as you read, and we'll revisit SS at the end of the brick.

GO TO CONCLUSION ↓

What Is Essential Thrombocythemia?

Essential thrombocythemia (ET) is a hematopoietic neoplasm characterized by massive platelet production. It is a type of **myeloproliferative neoplasm**, so it is part of a cluster of disorders in which the bone marrow produces large numbers of neoplastic, mature myeloid cells that end up spilling out into the blood. Patients usually have a more chronic and slower development of disease than in acute leukemias.

In most myeloprolife[illegible]ll line dominates over the others and is present in large numbers in the bone marrow and blood (although other lines may also increase). The **megakaryocytes** proliferate most in ET. Megakaryocytes are the precursors of platelets in the bone marrow, which are then released into the blood. So, in ET, there is a massive increase in platelet number (thrombocytosis). This can cause strokes and thrombosis.

Essential thrombocythemia is most common in older adult patients, but it also has a second, smaller peak of incidence in younger women. The annual incidence is approximately 1 to 2 cases per 100,000 people, similar to that of the other myeloproliferative neoplasms.

In essential thrombocythemia, what type of myeloid cell proliferates the most?

How Does a Patient With Essential Thrombocythemia Present (LO #1)?

Many patients with E[illegible]d when thrombocytosis (elevated platelet count) is found on a complete blood count (CBC) done for other reasons.

When symptoms are present, they fall into several groups:

- **Nonspecific vascular symptoms** such as headache, dizziness, syncope, and chest pain occur because of microthrombus formation or "sludginess" (slow flow) of the blood resulting from the massive number of platelets present. Some patients have episodic burning pain and redness in the distal extremities, a condition known as erythromelalgia (which is also seen in another myeloproliferative neoplasm, polycythemia vera).
- Less common are symptoms caused by **large thrombus formation**, leading to events such as stroke, deep vein thrombosis, or myocardial infarction caused by the abundance of platelets present.
- Least common are symptoms resulting from **excessive bleeding**. Why would too many platelets cause increased bleeding? This is because of **decreased platelet function,** sometimes from diminished function of von Willebrand factor, a platelet-associated protein necessary for forming blood clots. Pregnant patients with ET also have an increased risk of first trimester fetal loss.

What is the underlying mechanism causing erythromelalgia in a patient

What Is the Pathophysiology of Essential Thrombocythemia (LO #2)?

No environmental risk factors for ET have been clearly identified. The disease is caused by acquired or hereditary gene mutations.

Genetics

Although most cases of ET appear to be sporadic, families with an increased incidence have been described. Approximately 50% of cases of ET are associated with activating mutations in the *JAK2* (Janus kinase 2) gene. The ***JAK2* gene** codes for a tyrosine kinase that stimulates growth-promoting genes by phosphorylating JAK/signal transducer and activator of transcription (JAK/STAT) proteins. In hematopoietic stem cells, the JAK/STAT signaling pathway mediates the process of cell growth.

In JAK2-negative cases, there can be other mutations, such as calreticulin (*CALR*) (~30% of cases) or *MPL* (thrombopoietin receptor) (5–10% of cases).

What gene is commonly mutated in essential thrombocythemia?

Pathology

In ET, the abnormal growth of **megakaryocytes** in the bone marrow pushes many platelets into the peripheral blood.

Blood Smear. In ET, the main abnormality in the peripheral blood is the marked increase in the number of **platelets**. The platelets may range in size from tiny to gigantic (Figure 1). This range in different platelet sizes is called **platelet anisocytosis**. Other blood cells typically appear normal.

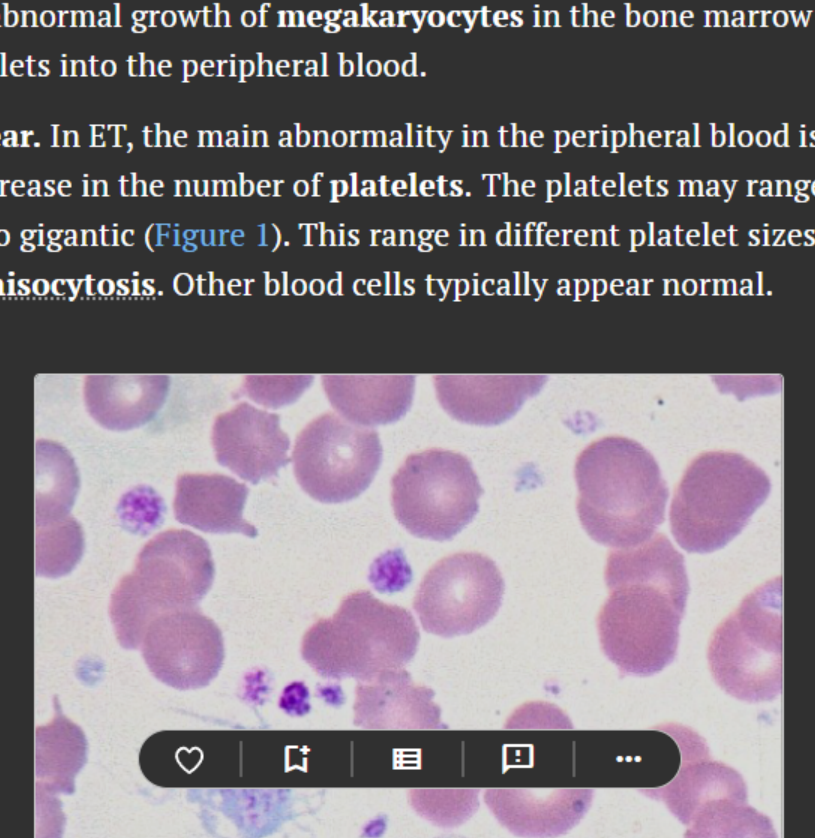

Figure 1

What is the main blood smear finding in ET?

Bone Marrow. The bone marrow in ET is typically normocellular. The unique thing about the bone marrow in ET is the massive proliferation of **megakaryocytes**, whi[illegible]ughout the bone marrow (Figure 2).

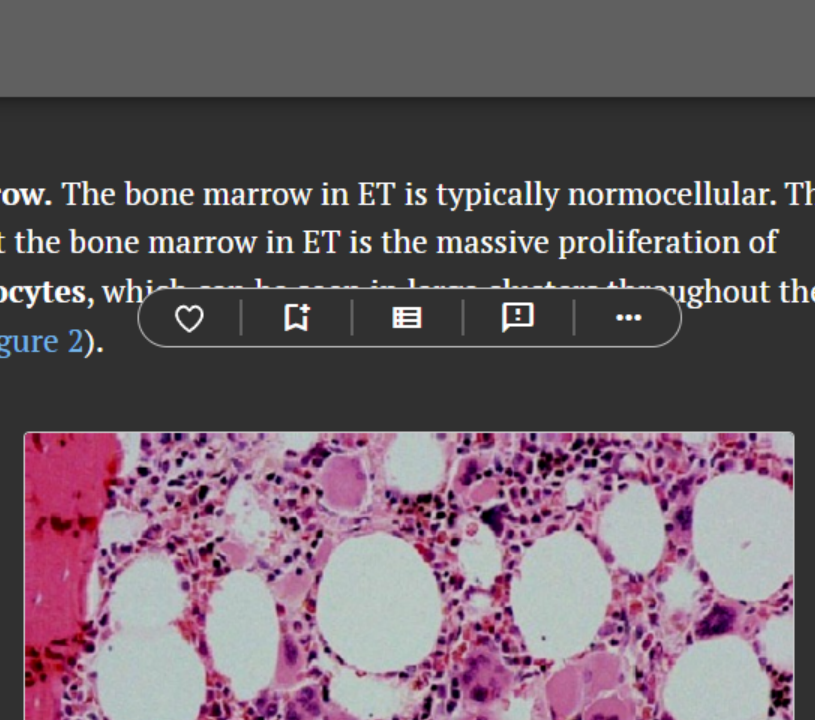

Figure 2

At the end stages of the disease, the bone marrow can become fibrotic, similar to findings at the end stages of the other myeloproliferative neoplasms.

Complications

In 1% to 2% of patients, ET may evolve into other hematopoietic neoplasms, including acute myeloid leukemia. This transformation is less common than in the other myeloproliferative neoplasms.

As mentioned above, ET can also lead to thrombus formation (causing myocardial infarctio[illegible]well as excess bleeding.

What bone marrow findings are present in essential thrombocythemia?

How Do We Diagnose Essential Thrombocythemia (LO #3)?

It's important to remember that not all patients with thrombocytosis have essential thrombocythemia! A high platelet count is pretty common, but ET is not.

When the platelet count is high for reasons other than ET, it is termed **secondary thrombocytosis.** Because this is much more common than ET is, we must first rule out the more common causes of secondary thrombocytosis before doing the genetic testing to confirm an ET diagnosis.

Excluding Seconda[illegible]

There are many conditions that can cause a secondary, or reactive, thrombocytosis. These include infections, surgery, malignancies, chronic inflammatory diseases, splenectomy, and anemia, especially iron deficiency anemia. These should be excluded by a history, physical, and CBC before considering the diagnosis of ET.

Which conditions can cause secondary thrombocytosis?

CLINICAL CORRELATION

An additional cause of secondary thrombocytosis would be pseudothrombocytosis, meaning the platelet count is artifactually elevated. This may occur in conditions that cause other cells to be counted as platelets, such as WBC fragments (in s[illegible]ic anemias), and microspherocytes (i[illegible]

Excluding Other Hematopoietic Neoplasias

A number of other types of hematopoietic neoplasias can cause thrombocytosis. These include other myeloproliferative neoplasms (polycythemia vera, primary myelofibrosis, chronic myeloid leukemia), myelodysplastic syndromes, and acute myeloid leukemia. This means most of the myeloid neoplasias, both acute and chronic. Most will be identifiable by other CBC or clinical findings.

Diagnosing Essential Thrombocythemia

Once secondary causes have been ruled out, we can consider ET.

Physical Exam. Like other myeloproliferative neoplasms, there can be splenomegaly (25%–50% of patients) caused by extramedullary hematopoiesis. Skin findings include livedo reticularis (a mottled, net-like purple rash caused by blood vessel narrowing) and erythromelalgia (burning and redness in the feet, hands, arms, face, and ears).

Peripheral Blood. The CBC will show **increased platelet number** (>400,000/mm^3) and, often, anisocytosis as shown above. Increased platelet size can be quantified by ordering a mean platelet volume. Increased volume suggests ET as opposed to secondary thrombocytosis, and this can be important in the differential diagnosis if the history is not definitive. There might also be findings seen in other myeloproliferative neoplasms, such as normocytic anemia, teardrop-shaped RBCs, nucleated RBCs, and immature granulocytes.

Bone Marrow Biopsy. If no secondary cause of thrombocytosis can be identified, a primary bone marrow biopsy is usually done to identify bone marrow neoplasia. **Dominant megakaryocyte proliferation** in the bone marrow suggests ET, but it is not always present.

Genetic Tests. After excluding secondary thrombocytosis, cells should be tested for characteristic genetic mutations such as those in *JAK2*, seen in half the cases of ET. The diagnosis is more challenging if this test is negative; typically, subsequent testing for *CALR* and *MPL* mutations is performed. Because ET doesn't have any classic diagnostic features (eg, the Philadelphia chromosome in chronic myeloid leukemia), it then becomes largely a diagnosis of exclusion. That is, when secondary thrombocytosis, the other myeloproliferative neoplasms, and acute hematopoietic malignancies have been ruled out completely, a diagnosis of ET can then be made.

How Do We Treat Essential Thrombocythemia (LO #4)?

There is no cure for ET, but medications can help manage symptoms. As one of the myeloproliferative neoplasms, ET has hallmark differences from PV, but they share many traits and their management with medication shares some similarities.

Recall from the brick on Polycythemia Vera that we discussed aspirin, hydroxyurea, and Pegylated interferon alfa (IFNa); these medications, their place in therapy, adverse effects, and contraindications apply in ET too.

- All patients with ET should receive low dose aspirin to reduce microvascular co[illegible]mplications.
- If aspirin alone d[illegible]if the patient is high-risk (> 60 years old OR history of thrombosis), then cytoreductive therapy with hydroxyurea or IFNa is added.
- In patients with PV and a history of thrombosis, or those who develop thrombosis, an anticoagulant is prescribed.

CLINICAL CORRELATION

Similar to PV, in ET about 50% of cases of ET are associated with activating mutations in the *JAK2* gene. This means that the JAK inhibitor ruxolitinib may be useful in ET patients as well.

What drug may be used to prevent clotting episodes in patients with essential thrombocythemia?

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to SS, what is the diagnosis? How do you describe her condition to her?

After the appropriate evaluation for secondary causes of this platelet elevation, you are convinced that SS has essential thrombocythemia. You explain that because of a mutation in a gene that leads to the production of growth-promoting substances, her bone marrow produces too many platelets. This can cause the headaches and visual blurring she is experiencing. You further explain that in some cases, excess clotting or excess bleeding can occur, and there are a few options for additional therapy. You prescribe low-dose aspirin and refer her to a hematologist to discuss the use of hydroxyurea.

Summary

- Essential thrombocythemia (ET) is a myeloproliferative neoplasm in which the bone marrow overproduces megakaryocytes, causing increased circulating platelets.
- Patients with ET are usually asymptomatic.
- When present, symptoms include headache, changes in vision, chest pain, or erythromelalgia.
- There may also be thrombosis and excessive bleeding.
- Approximately half of patients have a genetic mutation in *JAK2*.
- Peripheral blood smear may show an increased number of platelets and platelet anisocytosis.
- The bone marrow is hypercellular, with an increased number of megakaryocytes.
- Before diagnosing ET, first exclude causes of secondary thrombocytosis such as infection, recent surgery, or malignancy.
- If no secondary cause can be identified, a bone marrow biopsy is usually done, as well as genetic analysis for gene mutations.
- Treatment of symptoms should include low-dose aspirin.
- If a patient is at higher risk for serious complications, hydroxyurea may be added.

Review Questions

1. Which of the following presentations is least likely in a patient with essential thrombocythemia?

A. ○ Asymptomatic

B. ○ Bleeding gums

C. ○ Chest pain

D. ● Diarrhea

E. ○ Headache

Hide Explanation

The correct answer is diarrhea (D). Essential thrombocythemia (ET) does not usually present with gastroi[illegible]ell be asymptomatic (A) at diagnosis. They can have increased bleeding (B) resulting from reduced platelet function and increased clotting leading to coronary artery disease and chest pain (C). Patients with ET may also have microvascular symptoms such as headache (E).

2. Which of the following medications is most appropriate in a patient with essential thrombocythemia who has no history of thrombosis?

A. ○ Cyclophosphamide

B. ○ Lisinopril

C. ● Low-dose aspirin

D. ○ Metoprolol

E. ○ Tissue plasminogen activator

Hide Explanation

The correct answer is low-dose aspirin (C). Patients with essential thrombocythemia (ET) who are young and have no history of arterial or venous thrombi can be treated with low-dose aspirin. Hydroxyurea is added for patients with thrombi or who are at higher risk. Cyclophosphamide (A) is a chemotherapy drug not used for ET. Lisinopril (B) and metoprolol (D) are drugs used to control hypertension, not ET. A dose of tissue plasminogen activator (E) (a clot-busting drug) is used for some types of myocardial infarcti[illegible]

3. A 51-year-old patient with a history of multiple deep vein thromboses is being evaluated for a possible secondary cause. No recent surgery, infections, or cancers have been diagnosed. His complete blood count reveals marked thrombocytosis with numerous giant platelets seen on smear. Gene analysis shows a *JAK2* mutation. Which of the following findings is most likely on bone marrow biopsy?

A. ○ Abundant lymphoblasts

B. ● Normocellular marrow with many enlarged megakaryocytes

C. ○ Hypocellular marrow with fatty infiltration

D. ○ Normocellular marrow with numerous atypical macrophages

E. ○ Proliferation of monoclonal T cells

Hide Explanation

The correct answer is normocellular bone marrow with many enlarged megakaryocytes (B). Patients with essential thrombocythemia will likely show a normocellular bone marrow with an increased size and number of megakaryocytes. Abundant lymphoblasts (A) may be seen in many acute leukemias. Hypocellular bone marrow with fatty infiltration (C) is seen in aplastic anemia. A normocellular bone marrow with atypical macrophages (D) can be seen in Gaucher disease. Proliferation of mono[illegible]fter organ transplant.

4. A patient presents with clinical symptoms and bone marrow findings suspicious for essential thrombocythemia. *JAK2* testing is negative. Testing for what other gene mutation would be useful in this case?

A. ○ BCR::ABL1

B. ○ BRCA1

C. ● CALR

D. ○ FLT3

E. ○ TP53

Hide Explanation

The correct answer is CALR (C). In JAK2 negative cases, it is appropriate to test for the other MPN-associated mutations: CALR (~30% of cases) and MPL (5-10% of cases). BCR::ABL1 (A) mutations are seen in chronic myeloid leukemia. BRCA1 (B) mutations are not associated with myeloproliferative neoplasms. FLT3 (D) and TP53 (E) mutations are not associated with essential thrombocythemia but may be testing that is performed in acute myeloid leukemia (AML).

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of essential thrombocythemia.~~

02 ☑ ~~Explain~~[illegible]~~cythemia.~~

Hematopathology. 4th ed. CRC Press; 2021.
- Greer JP, et al, eds. *Wintrobe's Clinical Hematology.* 14th ed. Wolters Kluwer; 2018.
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04 ☑ ~~Outline the management plan of essential thrombocythemia.~~

Objective Confidence: 4/4